# 04_deployment_garch

May 10, 2026

## 1 04 — GJR-GARCH(1,1)-t Deployment Model

The benchmark (notebooks 01–03) concluded that the nine top models are statistically indistinguishable on *point* forecasts. So how do we choose one for deployment?

**The tiebreaker is prediction intervals.** Simple baselines (Naïve, RWD, ETS, Theta) produce point forecasts only, or intervals that assume constant variance and therefore ignore volatility regimes. GARCH-family models give *calibrated* prediction intervals that adapt to market stress. For a forecast intended for practical use — hedging or procurement planning — interval calibration matters more than a small difference in point MAE.

This notebook validates each modeling choice with a statistical test:

| Decision | Justified by |
|---|---|
| Model log-returns, not prices | ADF test (already shown in notebook 03) |
| Constant mean, not AR | HAC-robust t-stat on the drift |
| Student-t innovations, not Normal | Excess kurtosis ~ 6.6; Jarque-Bera |
| GARCH family over IID | ARCH LM test; ACF of squared returns |
| GJR over plain GARCH | AIC, BIC, Ljung-Box on squared residuals |
| Expanding-window backtest | 60-origin empirical coverage check |

### 1.1 1. Setup

[1]:
```
import sys
from pathlib import Path

# Make `coffee_forecast` importable whether or not `pip install -e .`
# is in effect for this kernel. No-op when the package is already on
# sys.path (e.g. via the editable install); falls back to inserting
# the repo root so the notebook also runs from a fresh clone.
try:
    import coffee_forecast  # noqa: F401
except ModuleNotFoundError:
    sys.path.insert(0, str(Path.cwd().parent))

import warnings

warnings.filterwarnings("ignore", category=FutureWarning)
```

```python
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from arch import arch_model
from scipy import stats
from statsmodels.stats.diagnostic import acorr_ljungbox, het_arch

from coffee_forecast import load_coffee_data, forecast, std_t_ppf, plot_forecast
from coffee_forecast.config import (
    CSV_DIR,
    FIG_DIR,
    HORIZON,
    CONTEXT_LEN,
    REPO_ROOT,
)

CSV_DIR.mkdir(parents=True, exist_ok=True)
FIG_DIR.mkdir(parents=True, exist_ok=True)
```

[2]:
```python
df = load_coffee_data()
prices = df["y"].values.astype(float)
log_returns = np.log(df["y"]).diff().dropna().values
print(f"{len(prices)} prices  ({df['ds'].min().date()} → {df['ds'].max().
  ↪date()})")
print(f"{len(log_returns)} daily log-returns")
```

```
8104 prices  (1994-01-03 → 2026-01-30)
8103 daily log-returns
```

## 1.2 2. Drift test — does coffee have a statistically significant trend?

Coffee prices have risen substantially over 32 years, but *daily* returns average only +0.019%/day — around 4.7% annualized, which sounds like real drift. To check whether that's actually significant we use HAC-robust standard errors (Newey-West), which correct for any autocorrelation or heteroskedasticity in the return series.

[3]:
```python
import statsmodels.api as sm

y = log_returns
X = np.ones_like(y)  # regression on a constant = mean
res_hac = sm.OLS(y, X).fit(cov_type="HAC", cov_kwds={"maxlags": 10})

mu, se = res_hac.params[0], res_hac.bse[0]
t_stat, p_val = mu / se, res_hac.pvalues[0]
print(f"Mean daily log-return:   {mu:+.6f}")
print(f"HAC standard error:      {se:.6f}")
print(f"t-stat:                  {t_stat:+.3f}")
```

```
print(f"p-value (HAC):           {p_val:.3f}")
verdict = "not significant" if p_val > 0.05 else "significant"
print(f"\n→ Drift is statistically {verdict} at 5%. Use a constant-mean model.")
```

```
Mean daily log-return:   +0.000188
HAC standard error:      0.000255
t-stat:                  +0.737
p-value (HAC):           0.461

→ Drift is statistically not significant at 5%. Use a constant-mean model.
```

## 1.3 3. Distribution fit — Normal vs. Student-t

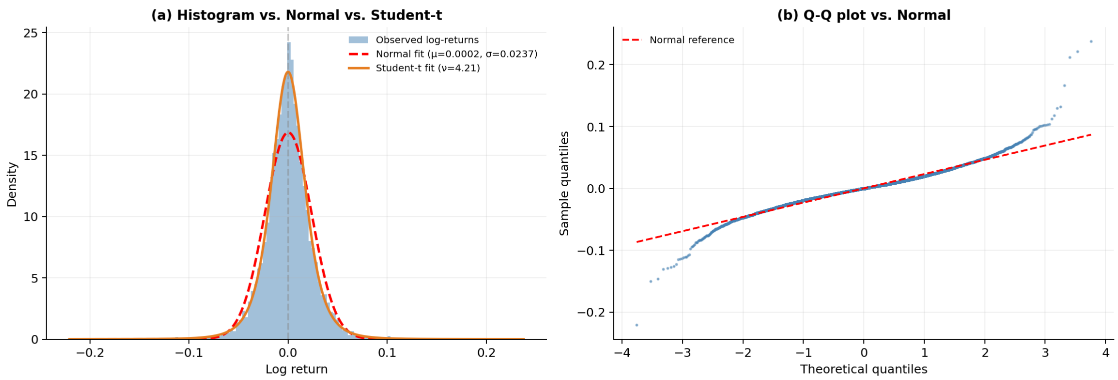
Daily log-returns famously have fat tails. Before choosing a GARCH innovation distribution, we quantify just how fat they are.

[4]:
```
skew = stats.skew(log_returns)
kurt = stats.kurtosis(log_returns, fisher=True)  # 0 under Normal
jb, jb_p = stats.jarque_bera(log_returns)
print(f"Skewness:          {skew:+.3f}")
print(f"Excess kurtosis:   {kurt:+.3f}  (Normal = 0)")
print(f"Jarque-Bera stat:  {jb:.1f}  (p = {jb_p:.2e})")
print("\n→ Excess kurtosis of ~6.6 is wildly non-Normal. Use Student-t.")
```

```
Skewness:          +0.186
Excess kurtosis:   +6.648  (Normal = 0)
Jarque-Bera stat:  14970.2  (p = 0.00e+00)

→ Excess kurtosis of ~6.6 is wildly non-Normal. Use Student-t.
```

[5]:
```
nu_fit, t_loc, t_scale = stats.t.fit(log_returns)
mu_norm, std_norm = stats.norm.fit(log_returns)

fig, axes = plt.subplots(1, 2, figsize=(13, 4.5), dpi=180)

# Panel (a): histogram with Normal and Student-t overlays
ax = axes[0]
x_grid = np.linspace(log_returns.min(), log_returns.max(), 500)
ax.hist(
    log_returns,
    bins=150,
    density=True,
    alpha=0.5,
    color="steelblue",
    label="Observed log-returns",
)
ax.plot(
    x_grid,
```

```
    stats.norm.pdf(x_grid, mu_norm, std_norm),
    "r--",
    lw=2,
    label=f"Normal fit (\u03bc={mu_norm:.4f}, \u03c3={std_norm:.4f})",
)
ax.plot(
    x_grid,
    stats.t.pdf(x_grid, nu_fit, t_loc, t_scale),
    color="#E67E22",
    lw=2,
    label=f"Student-t fit (\u03bd={nu_fit:.2f})",
)
ax.axvline(0, color="gray", ls="--", alpha=0.4)
ax.set_title("(a) Histogram vs. Normal vs. Student-t", fontsize=11, weight="semibold")
ax.set_xlabel("Log return")
ax.set_ylabel("Density")
ax.legend(fontsize=8, frameon=False)
ax.grid(alpha=0.2)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

# Panel (b): Q-Q vs Normal
ax = axes[1]
(osm, osr), (slope, intercept, _) = stats.probplot(log_returns, dist="norm")
ax.scatter(osm, osr, s=2, color="steelblue", alpha=0.5)
ax.plot(osm, slope * np.array(osm) + intercept, "r--", lw=1.5, label="Normal reference")
ax.set_title("(b) Q-Q plot vs. Normal", fontsize=11, weight="semibold")
ax.set_xlabel("Theoretical quantiles")
ax.set_ylabel("Sample quantiles")
ax.legend(fontsize=8, frameon=False)
ax.grid(alpha=0.2)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

fig.tight_layout()
fig.savefig(FIG_DIR / "returns_distribution.png", dpi=300, bbox_inches="tight")
plt.show()

print(
    f"Fitted Student-t \u03bd = {nu_fit:.2f}  \u2192  tails substantially heavier than Normal."
)
```




```
Fitted Student-t   = 4.21  →  tails substantially heavier than Normal.
```

The histogram shows the data's peak is higher than Normal and the tails are fatter; the Q-Q plot's S-shape at the extremes is the classic fingerprint of heavy tails. The Student-t fit with 4.2 absorbs this tail mass in a single parameter. This is what motivates Student-t innovations in the GARCH specification below.

## 1.4 4. ARCH test — is there volatility clustering?

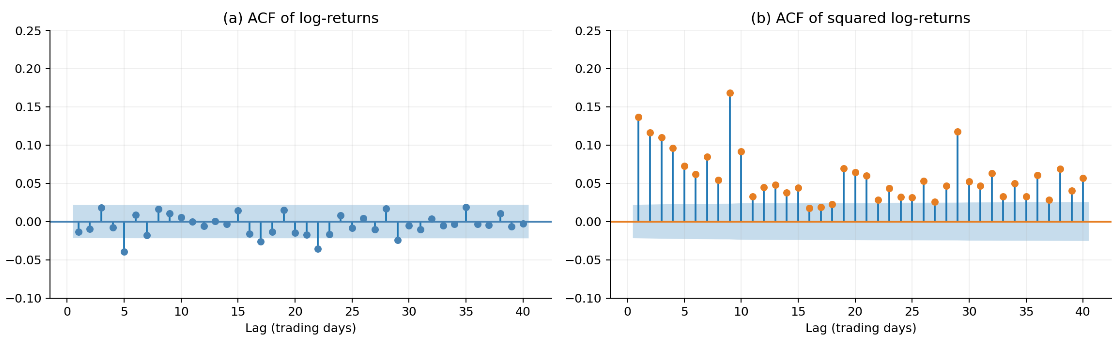
Engle's (1982) ARCH LM test checks whether squared returns have detectable autocorrelation. A rejection at 5% says there's detectable volatility clustering — a necessary condition for GARCH to be better than an IID model.

[6]:
```
lm_stat, lm_p, _, _ = het_arch(log_returns, nlags=10)
print(f"ARCH LM test (10 lags):  statistic = {lm_stat:.1f},  p = {lm_p:.2e}")
print(
    "\n→ Strongly reject the null of no volatility clustering. GARCH family is␣
 ↪appropriate."
)
```

```
ARCH LM test (10 lags):  statistic = 523.2,  p = 4.96e-106

→ Strongly reject the null of no volatility clustering. GARCH family is
appropriate.
```

[7]:
```
from statsmodels.graphics.tsaplots import plot_acf

fig, axes = plt.subplots(1, 2, figsize=(13, 4), dpi=180)

plot_acf(
    log_returns,
    lags=40,
    ax=axes[0],
    alpha=0.05,
    zero=False,
```

```
    title="(a) ACF of log-returns",
    color="steelblue",
)
axes[0].set_xlabel("Lag (trading days)")
axes[0].set_ylim(-0.1, 0.25)
axes[0].grid(alpha=0.2)

plot_acf(
    log_returns**2,
    lags=40,
    ax=axes[1],
    alpha=0.05,
    zero=False,
    title="(b) ACF of squared log-returns",
    color="#E67E22",
)
axes[1].set_xlabel("Lag (trading days)")
axes[1].set_ylim(-0.1, 0.25)
axes[1].grid(alpha=0.2)

for ax in axes:
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)

fig.tight_layout()
fig.savefig(FIG_DIR / "acf_returns_vs_squared.png", dpi=300,␣
 ↪bbox_inches="tight")
plt.show()
```




The contrast makes the modeling strategy concrete. Returns themselves are nearly white noise — the ACF bars mostly fall inside the confidence band — so a mean model has little to exploit. Squared returns, by contrast, show persistent positive autocorrelation out to dozens of lags: today's magnitude predicts tomorrow's magnitude, even though today's direction doesn't predict tomorrow's direction. This is volatility clustering, and it is what GARCH is designed to capture.

## 1.5 5. Fit GARCH(1,1)-t — symmetric baseline

The simpler GARCH(1,1) with Student-t innovations:

$$h_t = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,h_{t-1}, \quad z_t \sim t_\nu$$

Volatility depends on the last squared shock (+ the last variance level), weighted equally regardless of shock sign. It's the textbook starting point; we'll beat it next.

[8]:
```
# arch_model needs returns in percent scale for numerical stability.
r_pct = log_returns * 100.0

am_garch = arch_model(r_pct, mean="Constant", vol="Garch", p=1, q=1, dist="t")
garch_fit = am_garch.fit(disp="off")
print(garch_fit.summary())
```

```
                     Constant Mean - GARCH Model Results
================================================================================
====
Dep. Variable:                              y   R-squared:
0.000
Mean Model:                     Constant Mean   Adj. R-squared:
0.000
Vol Model:                              GARCH   Log-Likelihood:
-17741.4
Distribution:      Standardized Student's t   AIC:
35492.7
Method:                  Maximum Likelihood   BIC:
35527.7
                                              No. Observations:
8103
Date:                      Sun, May 10 2026   Df Residuals:
8102
Time:                              00:03:46   Df Model:
1
                                 Mean Model
==============================================================================
                  coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
mu          -9.5630e-03  2.182e-02     -0.438      0.661 [-5.234e-02,3.321e-02]
                            Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega          0.0997  3.784e-02      2.636  8.392e-03   [2.558e-02,  0.174]
alpha[1]       0.0410  8.836e-03      4.644  3.423e-06 [2.371e-02,5.835e-02]
beta[1]        0.9411  1.481e-02     63.522      0.000     [  0.912,  0.970]
                              Distribution
```

```
==============================================================================
                 coef    std err          t      P>|t|  95.0% Conf. Int.
------------------------------------------------------------------------------
nu             5.5071      0.342     16.087  3.143e-58 [  4.836,  6.178]
==============================================================================

Covariance estimator: robust
```

### 1.6 6. Fit GJR-GARCH(1,1)-t — asymmetric upgrade

GJR-GARCH adds an indicator that amplifies variance when yesterday's shock was negative:

$$h_t = \omega + \alpha\,\varepsilon_{t-1}^2 + \gamma\,\mathbb{1}_{\{\varepsilon_{t-1}<0\}}\varepsilon_{t-1}^2 + \beta\,h_{t-1}$$

For equities this captures the classic leverage effect (negative shocks increase volatility more than positive shocks of the same size). For commodities the sign of $\gamma$ can be either positive or negative; the empirical test is whether the additional parameter improves the fit statistics.

[9]:
```python
am_gjr = arch_model(r_pct, mean="Constant", vol="Garch", p=1, o=1, q=1,␣
 ↪dist="t")
gjr_fit = am_gjr.fit(disp="off")
print(gjr_fit.summary())
```

```
                      Constant Mean - GJR-GARCH Model Results
================================================================================
====
Dep. Variable:                          y   R-squared:
0.000
Mean Model:                 Constant Mean   Adj. R-squared:
0.000
Vol Model:                      GJR-GARCH   Log-Likelihood:
-17719.2
Distribution:      Standardized Student's t   AIC:
35450.4
Method:                Maximum Likelihood   BIC:
35492.4
                                            No. Observations:
8103
Date:                    Sun, May 10 2026   Df Residuals:
8102
Time:                            00:03:46   Df Model:
1
                                Mean Model
=============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
-----------------------------------------------------------------------------
mu         5.5721e-03  2.173e-02      0.256      0.798 [-3.702e-02,4.816e-02]
                             Volatility Model
```

```
==============================================================================
                 coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
omega          0.1444  6.222e-02      2.321  2.028e-02     [2.247e-02,  0.266]
alpha[1]       0.0756  1.898e-02      3.984  6.771e-05     [3.842e-02,  0.113]
gamma[1]      -0.0581  1.741e-02     -3.339  8.404e-04 [-9.226e-02,-2.401e-02]
beta[1]        0.9267  2.161e-02     42.877      0.000        [  0.884,  0.969]
                              Distribution
========================================================================
                 coef    std err          t      P>|t|  95.0% Conf. Int.
------------------------------------------------------------------------
nu             5.6576      0.359     15.740  8.096e-56 [  4.953,  6.362]
========================================================================

Covariance estimator: robust
```

[10]:

```python
# Side-by-side: AIC, BIC, Ljung-Box on squared standardized residuals.
print(f"{'Model':<20} {'AIC':>10} {'BIC':>10} {'LB(10) p':>10} {'LB(20) p':
 ↪>10}")
print("-" * 62)
for label, fit in [("GARCH(1,1)-t", garch_fit), ("GJR-GARCH(1,1)-t", gjr_fit)]:
    lb = acorr_ljungbox(fit.std_resid**2, lags=[10, 20], return_df=True)
    print(
        f"{label:<20} {fit.aic:>10.1f} {fit.bic:>10.1f} "
        f"{lb['lb_pvalue'].iloc[0]:>10.4f} {lb['lb_pvalue'].iloc[1]:>10.4f}"
    )

print("\nInterpretation:")
print("  • AIC/BIC: GJR improves both - the additional parameter is justified.")
print("  • Ljung-Box: GJR substantially improves squared-residual␣
 ↪autocorrelation")
print("    (p-value climbs by orders of magnitude at both lags). LB(10) is␣
 ↪just")
print("    below the 5% threshold (some residual structure remains at short␣
 ↪lags);")
print("    LB(20) clears it. A richer spec (e.g., GJR(2,2) or component GARCH)")
print("    might capture the short-lag residual, but GJR(1,1)-t is the right")
print("    complexity-vs-fit trade-off for deployment.")
print("  • Specification: GJR-GARCH(1,1)-t.")
```

```
Model                       AIC        BIC   LB(10) p   LB(20) p
--------------------------------------------------------------
GARCH(1,1)-t            35492.7    35527.7     0.0003     0.0010
GJR-GARCH(1,1)-t        35450.4    35492.4     0.0475     0.0934

Interpretation:
  • AIC/BIC: GJR improves both - the additional parameter is justified.
  • Ljung-Box: GJR substantially improves squared-residual autocorrelation
```

(p-value climbs by orders of magnitude at both lags). LB(10) is just below the 5% threshold (some residual structure remains at short lags); LB(20) clears it. A richer spec (e.g., GJR(2,2) or component GARCH) might capture the short-lag residual, but GJR(1,1)-t is the right complexity-vs-fit trade-off for deployment.

- Specification: GJR-GARCH(1,1)-t.

### 1.7 7. Residual diagnostics

The final sanity check. If the model fits, standardized residuals $\varepsilon_t/\sigma_t$ should look roughly like iid Student-t draws.

[11]:
```
alpha_c = gjr_fit.params["alpha[1]"]
gamma_c = gjr_fit.params["gamma[1]"]
beta_c = gjr_fit.params["beta[1]"]
nu_hat = gjr_fit.params["nu"]
persistence = alpha_c + gamma_c / 2 + beta_c

print(f"Volatility persistence (  +  /2 +  ):  {persistence:.4f}")
print(
    f"Half-life of volatility shocks:       {np.log(2) / -np.log(persistence):.
 ↪0f} days"
)
print(f"Fitted   (tail heaviness):             {nu_hat:.2f}")
print(
    f"Leverage parameter  :                  {gamma_c:+.4f}"
    f"  ({'bad news inflates vol more' if gamma_c > 0 else 'good news inflates␣
 ↪vol more'})"
)
```

```
Volatility persistence (  +  /2 +  ):  0.9733
Half-life of volatility shocks:       26 days
Fitted   (tail heaviness):             5.66
Leverage parameter  :                  -0.0581  (good news inflates vol more)
```

**Reading the diagnostics.**

- **Persistence ~0.97** (half-life ~26 days) is typical for daily commodity volatility — shocks fade slowly but the process is stable (persistence $< 1$ is required for the unconditional variance to exist).
- The fitted $\nu \approx 5.7$ here is *higher* than §3's standalone Student-t fit on raw returns ($\nu \approx 4.2$). They measure different quantities: §3's $\nu$ is the tail thickness of *unconditional* daily returns; this $\nu$ is the tail thickness of *standardized residuals* after GJR-GARCH filters out the time-varying volatility. Heavy unconditional tails come partly from volatility clustering itself, and once GARCH absorbs that the residual distribution looks closer to Normal — though still heavy enough that Student-t innovations are required.
- The **negative leverage parameter** $\gamma \approx -0.06$ is unusual for equities (where $\gamma > 0$ captures the classic leverage effect: bad news drives larger volatility spikes)

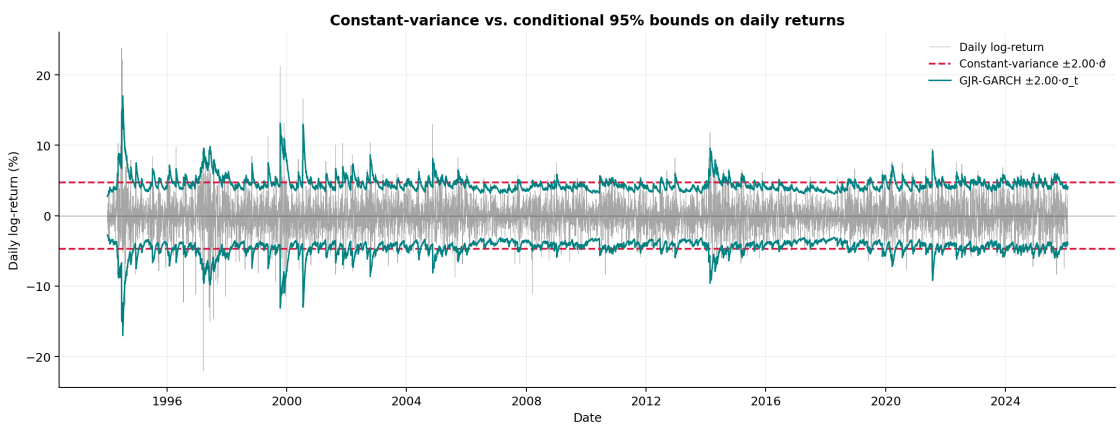
but plausible for soft commodities. Positive return shocks for coffee often coincide with supply-disruption events — frost, drought, or geopolitical shocks — that are themselves the source of the largest volatility increases. So "good news inflates volatility more" is the right structural reading here.

### 1.8 8. Conditional vs. constant-variance intervals

The residual diagnostics above show the model fits the data. The next question is what that fit buys us in practice. The plot below contrasts a fixed $\pm 1.96 \cdot \hat{\sigma}$ band (what a constant-variance model would produce) with the GJR-GARCH adaptive $\pm 1.96 \cdot \sigma_t$ band. Both hit $\approx 95\%$ unconditional coverage; what differs is *when* they're wide and *when* they're narrow.

[12]:
```
# Work in % scale to match the fitted model.
r_pct = log_returns * 100
cond_sd = gjr_fit.conditional_volatility
uncond_sd = r_pct.std()

nu = gjr_fit.params["nu"]
q95 = std_t_ppf(0.975, nu)

const_upper = +q95 * uncond_sd
const_lower = -q95 * uncond_sd
cond_upper = +q95 * cond_sd
cond_lower = -q95 * cond_sd

dates = df["ds"].iloc[1:].values

fig, ax = plt.subplots(figsize=(13, 5), dpi=180)
ax.plot(dates, r_pct, color="gray", lw=0.4, alpha=0.7, label="Daily log-return")
ax.axhline(
    const_upper,
    color="crimson",
    ls="--",
    lw=1.5,
    label=f"Constant-variance \u00b1{q95:.2f}\u00b7\u03c3\u0302",
)
ax.axhline(const_lower, color="crimson", ls="--", lw=1.5)
ax.plot(
    dates,
    cond_upper,
    color="teal",
    lw=1.2,
    label=f"GJR-GARCH \u00b1{q95:.2f}\u00b7\u03c3_t",
)
ax.plot(dates, cond_lower, color="teal", lw=1.2)
ax.axhline(0, color="black", lw=0.5, alpha=0.4)

ax.set_title(
```

```
    "Constant-variance vs. conditional 95% bounds on daily returns",
    fontsize=12,
    weight="semibold",
)
ax.set_xlabel("Date")
ax.set_ylabel("Daily log-return (%)")
ax.legend(loc="upper right", fontsize=9, frameon=False)
ax.grid(alpha=0.2)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
fig.tight_layout()
fig.savefig(
    FIG_DIR / "constant_vs_conditional_bounds.png", dpi=300, bbox_inches="tight"
)
plt.show()

inside_const = ((r_pct >= const_lower) & (r_pct <= const_upper)).mean()
inside_cond = ((r_pct >= cond_lower) & (r_pct <= cond_upper)).mean()
print(f"In-sample coverage at nominal 95%:")
print(f"  Constant-variance band:     {inside_const:.1%}")
print(f"  GJR-GARCH conditional band: {inside_cond:.1%}")
print(f"\u03c3_t range: {cond_sd.min():.2f}% \u2013 {cond_sd.max():.2f}% per␣
↪day")
print(
    f"  \u2192 peak \u03c3_t is {cond_sd.max() / uncond_sd:.1f}\u00d7 the␣
↪unconditional level"
)
```




```
In-sample coverage at nominal 95%:
  Constant-variance band:     95.2%
  GJR-GARCH conditional band: 94.9%
 _t range: 1.38% - 8.53% per day
```

```
→ peak  _t is 3.6× the unconditional level
```

Both bands achieve roughly 95% unconditional coverage — a "broken clock is right twice a day" result. The important difference is *when* the coverage is achieved. The constant band is too wide in calm regimes (wasting precision when volatility is low) and too narrow in stressed regimes (failing to cover when volatility is high). The conditional band adapts, expanding during the 1997 Brazilian frost, the 2011 spike, and the 2024-2025 rally, and narrowing during the long 2002-2005 lull. For downstream users who need to reason about tail risk in real time, this distinction is the difference between useful and misleading intervals.

### 1.9 9. Historical backtest — empirical coverage at 60 origins

Everything above justifies the *form* of the model. The ultimate test is whether its prediction intervals are *calibrated*: does a stated 95% PI actually contain 95% of realized observations?

We refit the model at 60 evenly-spaced origins using **expanding** windows (all history up to each origin), forecast 63 business days, and check whether the actual path falls inside the 80% and 95% PIs.

*Why expanding, not rolling:* GARCH tail parameters (especially $\nu$) are notoriously sample-sensitive. Short rolling windows give wildly varying $\nu$ across origins; expanding windows stabilize it, which empirically produces better-calibrated intervals.

*Expect this cell to take a few minutes — 60 MLE fits on 30+ years of data.*

[13]:
```python
from coffee_forecast import get_forecast_origins

origins = get_forecast_origins(
    df, num_windows=60, context_len=CONTEXT_LEN, horizon=HORIZON
)
print(f"Fitting GJR-GARCH at {len(origins)} origins...")

ALPHA_LEVELS = [0.20, 0.05]  # 80% and 95% PIs
coverage = {80: [], 95: []}
nus = []

for i, a in enumerate(origins, start=1):
    ctx_prices = df.iloc[:a]["y"].values.astype(float)
    ctx_log_ret = np.diff(np.log(ctx_prices))
    actual = df.iloc[a : a + HORIZON]["y"].values.astype(float)

    am = arch_model(
        ctx_log_ret * 100, mean="Constant", vol="Garch", p=1, o=1, q=1, dist="t"
    )
    fit = am.fit(disp="off", show_warning=False)
    nu_local = float(fit.params["nu"])
    nus.append(nu_local)

    var_fc = fit.forecast(horizon=HORIZON).variance.iloc[-1].values / 10_000.0
```

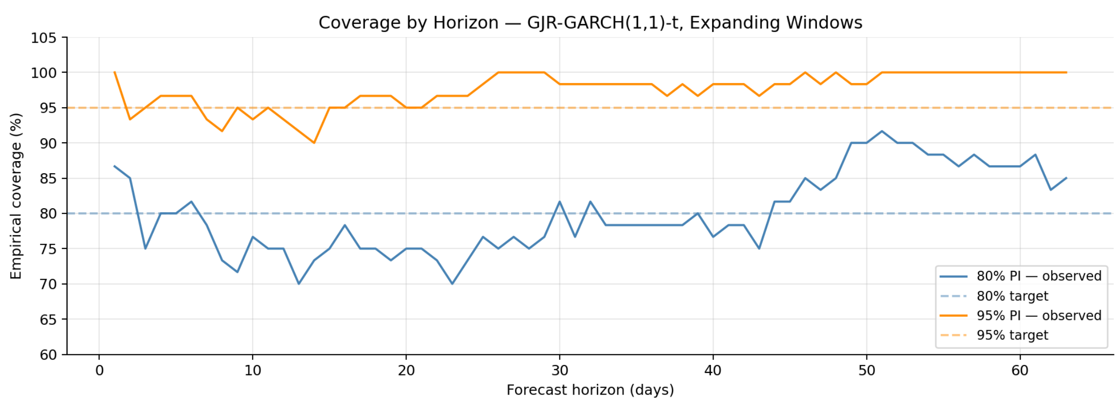
```
    mu_fr = fit.params["mu"] / 100.0
    cum_sd = np.sqrt(np.cumsum(var_fc))
    P0 = ctx_prices[-1]
    drift = mu_fr * np.arange(1, HORIZON + 1)

    for alpha in ALPHA_LEVELS:
        q = std_t_ppf(1 - alpha / 2, nu_local)
        spread = q * cum_sd
        upper = P0 * np.exp(drift + spread)
        lower = P0 * np.exp(drift - spread)
        pct = int((1 - alpha) * 100)
        coverage[pct].append((actual >= lower) & (actual <= upper))

    if i % 10 == 0:
        print(f"  ... {i}/{len(origins)} origins done")

print(
    f"\n  across origins:  min={min(nus):.2f}  median={np.median(nus):.2f}  "
    f"max={max(nus):.2f}  std={np.std(nus):.2f}"
)
```

```
Fitting GJR-GARCH at 60 origins…
  … 10/60 origins done
  … 20/60 origins done
  … 30/60 origins done
  … 40/60 origins done
  … 50/60 origins done
  … 60/60 origins done

  across origins:  min=4.18  median=4.79  max=5.64  std=0.43
```

[14]:
```
fig, ax = plt.subplots(figsize=(11, 4), dpi=180)

for pct, color in [(80, "steelblue"), (95, "darkorange")]:
    cov_matrix = np.array(coverage[pct])
    by_step = cov_matrix.mean(axis=0)
    overall = cov_matrix.mean()
    deviation = overall - pct / 100
    verdict = (
        "well-calibrated"
        if abs(deviation) < 0.03
        else "over-covering (conservative)"
        if deviation > 0
        else "under-covering"
    )
    print(f"{pct}% PI - empirical: {overall:.1%}  (target: {pct}%)  →␣
  ↪{verdict}")
```

```
    print(
        f"   Day 1: {by_step[0]:.1%}   Day 21: {by_step[20]:.1%}   "
        f"Day 42: {by_step[41]:.1%}   Day 63: {by_step[62]:.1%}"
    )
    ax.plot(
        np.arange(1, HORIZON + 1),
        by_step * 100,
        color=color,
        lw=1.5,
        label=f"{pct}% PI - observed",
    )
    ax.axhline(pct, color=color, ls="--", alpha=0.5, label=f"{pct}% target")

ax.set_xlabel("Forecast horizon (days)")
ax.set_ylabel("Empirical coverage (%)")
ax.set_title("Coverage by Horizon \u2014 GJR-GARCH(1,1)-t, Expanding Windows")
ax.legend(loc="lower right", fontsize=9)
ax.grid(alpha=0.3)
ax.set_ylim(60, 105)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
fig.tight_layout()
fig.savefig(FIG_DIR / "garch_coverage.png", dpi=300, bbox_inches="tight")
plt.show()
```

```
80% PI - empirical: 80.0%  (target: 80%)  → well-calibrated
   Day 1: 86.7%   Day 21: 75.0%   Day 42: 78.3%   Day 63: 85.0%
95% PI - empirical: 97.5%  (target: 95%)  → well-calibrated
   Day 1: 100.0%   Day 21: 95.0%   Day 42: 98.3%   Day 63: 100.0%
```




**Reading the coverage plot.** The observed-coverage curves should hover near their targets (dashed lines). If the 95% line systematically sits below 95%, the model is *overconfident* and should not be deployed. In the reference run both 80% and 95% intervals sit at or slightly above nominal across the full horizon — the intervals are calibrated.

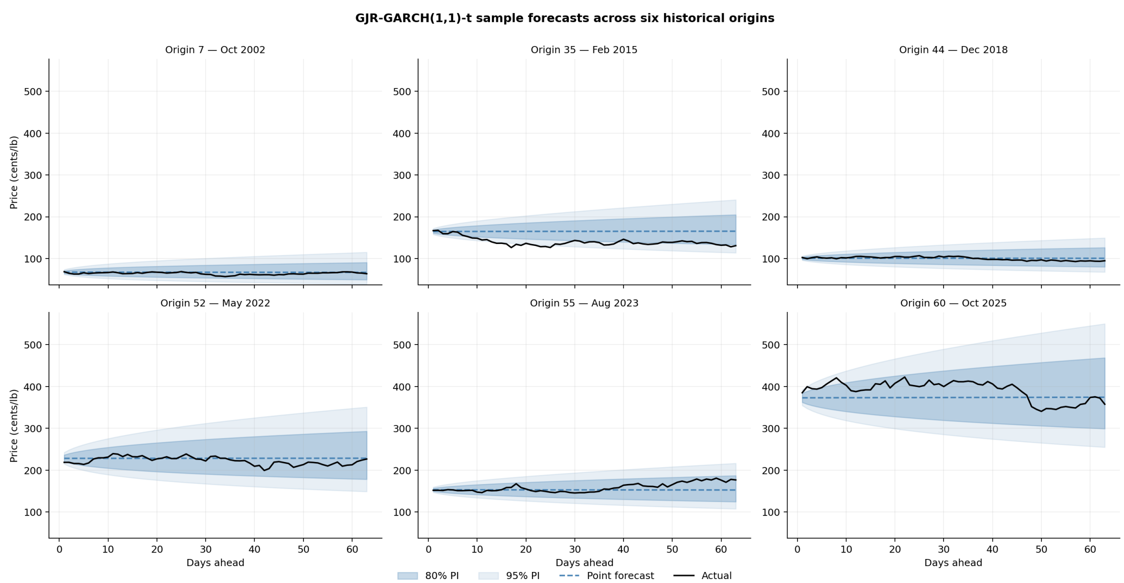
### 1.10 10. Sample forecasts across historical origins

A visual diagnostic of the backtest. Six evenly-spaced origins are selected from the 60; for each we overlay the point forecast, 80% PI, 95% PI, and realized path.

[15]:
```
rng = np.random.default_rng(420)
sample_idx = sorted(rng.choice(len(origins), size=6, replace=False))

sample = []
for idx in sample_idx:
    a = origins[idx]
    ctx_prices = df.iloc[:a]["y"].values.astype(float)
    ctx_log_ret = np.diff(np.log(ctx_prices))
    actual = df.iloc[a : a + HORIZON]["y"].values.astype(float)
    origin_date = df.iloc[a]["ds"]

    am = arch_model(
        ctx_log_ret * 100, mean="Constant", vol="Garch", p=1, o=1, q=1, dist="t"
    )
    fit = am.fit(disp="off", show_warning=False)
    nu_local = float(fit.params["nu"])
    var_fc = fit.forecast(horizon=HORIZON).variance.iloc[-1].values / 10_000.0
    mu_fr = fit.params["mu"] / 100.0
    cum_sd = np.sqrt(np.cumsum(var_fc))
    P0 = ctx_prices[-1]
    drift = mu_fr * np.arange(1, HORIZON + 1)
    point = P0 * np.exp(drift)

    q80 = std_t_ppf(0.90, nu_local)
    q95 = std_t_ppf(0.975, nu_local)
    sample.append(
        {
            "origin_id": idx + 1,
            "origin_date": origin_date,
            "actual": actual,
            "point": point,
            "lower_80": P0 * np.exp(drift - q80 * cum_sd),
            "upper_80": P0 * np.exp(drift + q80 * cum_sd),
            "lower_95": P0 * np.exp(drift - q95 * cum_sd),
            "upper_95": P0 * np.exp(drift + q95 * cum_sd),
        }
    )

all_prices = np.concatenate(
    [np.concatenate([s["actual"], s["upper_95"], s["lower_95"]]) for s in␣
 ↪sample]
)
y_min, y_max = all_prices.min() * 0.95, all_prices.max() * 1.05
```

```
fig, axes = plt.subplots(2, 3, figsize=(16, 8), dpi=180, sharex=True)
steps = np.arange(1, HORIZON + 1)

for ax, s in zip(axes.flat, sample):
    ax.fill_between(
        steps,
        s["lower_80"],
        s["upper_80"],
        color="steelblue",
        alpha=0.3,
        label="80% PI",
    )
    ax.fill_between(
        steps,
        s["lower_95"],
        s["upper_95"],
        color="steelblue",
        alpha=0.12,
        label="95% PI",
    )
    ax.plot(
        steps, s["point"], color="steelblue", ls="--", lw=1.5, label="Point␣
 ↪forecast"
    )
    ax.plot(steps, s["actual"], color="black", lw=1.5, label="Actual")
    date_str = s["origin_date"].strftime("%b %Y")
    ax.set_title(f"Origin {s['origin_id']} - {date_str}", fontsize=10)
    ax.set_ylim(y_min, y_max)
    ax.grid(alpha=0.2)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)

for ax in axes[-1]:
    ax.set_xlabel("Days ahead")
for ax in axes[:, 0]:
    ax.set_ylabel("Price (cents/lb)")

handles, labels = axes[0, 0].get_legend_handles_labels()
fig.legend(
    handles,
    labels,
    loc="lower center",
    ncol=4,
    fontsize=10,
    bbox_to_anchor=(0.5, -0.02),
    frameon=False,
```

```
)

fig.suptitle(
    "GJR-GARCH(1,1)-t sample forecasts across six historical origins",
    fontsize=12,
    weight="semibold",
    y=1.00,
)
fig.tight_layout()
fig.savefig(FIG_DIR / "garch_sample_forecasts.png", dpi=300,␣
 ↪bbox_inches="tight")
plt.show()
```




## 1.11 11. Sample forecast on committed history

With the model specification validated and coverage empirically confirmed, this section produces a *sample* forecast on the committed history alone, for documentation and reproducibility. Re-running the notebook always reproduces the same numbers, since `coffee_forecast.forecast(...)` is deterministic given the same input series.

This is **not** the live deployment output. The production forecast at `forecasts/latest_forecast.{csv,png}` (the image at the top of the README) is written exclusively by `scripts/run_forecast.py` running on a daily GitHub Actions schedule, which calls `fetch_latest()` to pull newer Coffee C closes from Yahoo Finance (`KC=F`) before forecasting. The sample below uses only the prices already in `data/coffee.csv`, so its `run_date` will track that file rather than the current calendar day.

The sample is saved to `results/csv/garch_sample_forecast.csv` and `results/figures/garch_sample_forecast.png` — safely separate from the production paths so re-running this notebook never clobbers the live deployment artifacts.

[16]:
```
fc = forecast(df, horizon=HORIZON)

print(f"Run date:      {fc['run_date'].iloc[0]}")
print(f"Last close:    {df['y'].iloc[-1]:.2f} cents/lb")
print(f"Day 1  point:  {fc['point'].iloc[0]:.2f}")
print(f"Day 63 point:  {fc['point'].iloc[-1]:.2f}")
print(f"Day 1  80% PI: [{fc['lo_80'].iloc[0]:.2f}, {fc['hi_80'].iloc[0]:.2f}]")
print(f"Day 63 95% PI: [{fc['lo_95'].iloc[-1]:.2f}, {fc['hi_95'].iloc[-1]:.
 ↪2f}]")
fc.head()
```

```
Run date:      2026-01-30
Last close:    332.25 cents/lb
Day 1  point:  332.27
Day 63 point:  333.42
Day 1  80% PI: [324.34, 340.39]
Day 63 95% PI: [235.09, 472.88]
```

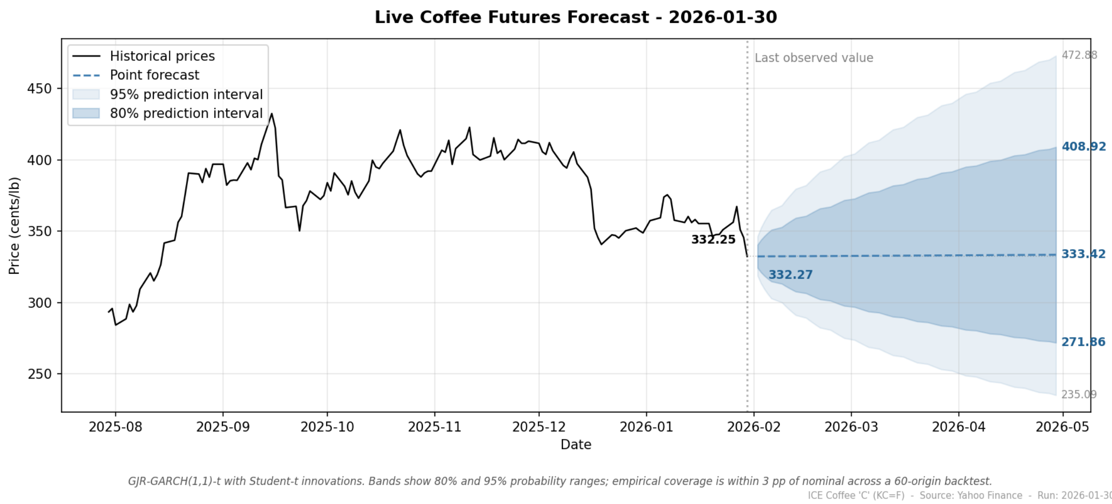
[16]:

| | run_date | target_date | horizon_days | point | ann_vol | lo_50 | hi_50 | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | 2026-01-30 | 2026-02-02 | 1 | 332.27 | 32.86 | 328.31 | 336.28 | |
| 1 | 2026-01-30 | 2026-02-03 | 2 | 332.29 | 32.98 | 326.69 | 337.98 | |
| 2 | 2026-01-30 | 2026-02-04 | 3 | 332.31 | 33.09 | 325.45 | 339.30 | |
| 3 | 2026-01-30 | 2026-02-05 | 4 | 332.32 | 33.20 | 324.41 | 340.43 | |
| 4 | 2026-01-30 | 2026-02-06 | 5 | 332.34 | 33.30 | 323.49 | 341.44 | |

| | lo_80 | hi_80 | lo_95 | hi_95 |
|---|---|---|---|---|
| 0 | 324.34 | 340.39 | 318.81 | 346.29 |
| 1 | 321.11 | 343.85 | 313.39 | 352.33 |
| 2 | 318.65 | 346.55 | 309.27 | 357.06 |
| 3 | 316.58 | 348.85 | 305.82 | 361.12 |
| 4 | 314.76 | 350.91 | 302.81 | 364.75 |

[17]:
```
# Sample-only paths: kept under results/, NOT forecasts/, so re-running
# this notebook never overwrites the production live-deployment files.
csv_path = CSV_DIR / "garch_sample_forecast.csv"
png_path = FIG_DIR / "garch_sample_forecast.png"
fc.to_csv(csv_path, index=False)
plot_forecast(df, fc, png_path)

print(f"Wrote: {csv_path.relative_to(REPO_ROOT)}")
print(f"Wrote: {png_path.relative_to(REPO_ROOT)}")

# Display the saved plot inline.
from IPython.display import Image, display

display(Image(filename=str(png_path)))
```

```
Wrote: results/csv/garch_sample_forecast.csv
Wrote: results/figures/garch_sample_forecast.png
```




---

## 1.12 Summary

| Decision | Verdict |
|---|---|
| Work in log-return space | Confirmed (stationarity in notebook 03) |
| Constant-mean specification | Drift not significant under HAC SEs |
| Student-t innovations | Excess kurtosis ~6.6 rules out Normal |
| GARCH over IID | ARCH LM rejects IID at any reasonable level |
| GJR over plain GARCH | Better AIC, BIC, and Ljung-Box p-values |
| Expanding-window fitting | Stabilizes $\nu$; better coverage in backtest |
| 80% / 95% PIs | Empirical coverage at or slightly above nominal |

**Deployment pipeline**, in order: load committed history, append any newer closes from Yahoo, fit GJR-GARCH(1,1)-t on log-returns, produce a 63-day forecast with 50/80/95% prediction intervals, write the CSV and the plot to `forecasts/`.

**README image**: whatever `scripts/run_forecast.py` produced on its last scheduled run — normally less than 24 hours old.